VERY AMPLENESS FROM SURJECTIVITY OF RESTRICTION TO ZERO DIMENSIONAL SUBSCHEMES OF DEGREE TWO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a complete reformulation and the author writes out both translations explicitly; nothing is referred to the reader.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-variant-separate-points-tangent-vectors is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Very ampleness from separating closed points and nonzero tangent vectors by sections. Same chapter, Lemma lemma-separate-points-tangent-vectors, invoked at proof lines 4399-4400. Supplies the conclusion once the two separation hypotheses are obtained.
- A nonzero tangent vector corresponds to a closed immersion of the spectrum of the dual numbers. Standard, used at proof lines 4404-4407. Supplies the degree two zero dimensional subscheme attached to a tangent vector.

Source reference. Original lemma 0E8S: sources/stacks-varieties/pinned-original-source.tex, source lines 4309–4309. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be an algebraically closed field. Let X be a proper k-scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Suppose that for every closed subscheme $Z \subset X$ of dimension 0 and degree 2 over k the map*

$$H^0(X, \mathcal{L}) \longrightarrow H^0(Z, \mathcal{L}|_Z)$$

is surjective. Then $\mathcal{L}$ is very ample on X over k.

Proof. This is a reformulation of Lemma 0E8S. Namely, given distinct closed points $x, y \in X$ taking $Z = x \cup y$ (viewed as closed subscheme) we get condition (1) of the lemma. And given a nonzero tangent vector $\theta \in T_{X/k,x}$ the morphism $\theta : \mathrm{Spec}(k[\epsilon]) \to X$ is a closed immersion. Setting $Z = \mathrm{Im}(\theta)$ we obtain condition (2) of the lemma. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.